NOTENEST | MODEL PRACTICE PAPER | DEMO

DSA Model Practice Paper

Data Structures and Algorithms - Demo paper - 2024 edition - 50 marks

> *Self-created practice paper for NoteNest. It is not an official university paper and is not taken from any real examination.*

Section A - Short answers (4 x 4 marks)

Q1. Define Big-O notation and state the complexity of binary search and of insertion at the head of a linked list. **[4 marks]**

Q2. Convert the infix expression (A + B) * C - D to postfix and evaluate it for A=2, B=3, C=4, D=5. **[4 marks]**

Q3. Differentiate between BFS and DFS with one application of each. **[4 marks]**

Q4. Why is the worst case of quick sort O(n^2)? Name one way to avoid it. **[4 marks]**

Section B - Long answers (2 x 17 marks)

Q5. Insert 50, 30, 70, 20, 40, 60, 80 into an empty BST. Draw the tree, give all three depth-first traversals and show how to delete 30. **[17 marks]**

Q6. Explain Dijkstra's algorithm. Apply it to a 5-node weighted graph of your choice and list the shortest distance to each node. State its time complexity with a binary heap. **[17 marks]**

Answer hints

- Q2: postfix is A B + C * D - ; the value is (2+3)*4-5 = 15.
- Q4: a bad pivot (smallest or largest element every time) gives unbalanced partitions; use a random or median-of-three pivot.
- Q5: inorder of the BST is 20 30 40 50 60 70 80.